1.

The ordering of shielding materials to provide maximum protection from a fast neutron source starting at a point closest to the source should be which of the following?

1. Lead, boron, plastic
2. Plastic, boron, lead
3. Boron, plastic, lead
4. Boron, lead, plastic
5. Plastic, lead, boron

2.

The immediate purpose of physical controls in radiation protection is to:

1. maintain personnel exposure within federal limits.
2. prevent inadvertent exposure.
3. keep hazards a secret.
4. protect sensitive equipment.
5. prevent authorized radiation exposure.

3.

The most significant radiation hazard associated with a PuBe source is:

1. alpha particles.
2. beta particles.
3. gamma photons.
4. neutrons.
5. neutrinos.

4.

An example of an engineering control to reduce EXTERNAL exposure to radiation is:

1. installation of fail-safe interlocks on doors or equipment.
2. restricting access to certain areas.
3. requiring the use of work permits.
4. requiring the use of respirators in airborne radioactivity areas.
5. use of glove box containments.

5.

The most effective ALARA program will be based on:

1. a good radiological control supervisor.
2. worker awareness of hazards in the work place.
3. a large radiological control staff.
4. a strong management commitment to radiation protection.
5. the size of the work force.

6.

Which of the following is a function of an effective ALARA program?

1. Train workers in methods to reduce exposure.
2. Increase the number of qualified workers to reduce individual worker dose.
3. Enable radiation workers to design shielding systems.
4. Minimize a company's legal liability in accidents.
5. Place total responsibility for ALARA upon the worker.

7.

An adequate Radiation Safety Program is the ultimate responsibility of the:

1. owner of the licensed facility.
2. operator of the source of radiation.
3. Radiation Safety Officer.
4. line supervisor.
5. radiation worker.

8.

In the transportation of radioactive materials, the PRIMARY means of achieving safety is:

1. effective use of time, distance, and shielding by everyone involved.
2. restricting transport to times of low traffic density.
3. proper approved packaging for the contents.
4. use of placard and labels that contain warnings.
5. proper procedures to prevent damage from rough handling.

9.

The range of a 2 MeV beta particle in soft tissue would be approximately which of the following?

1. 0.1
2. 1 cm
3. 2 cm
4. 3 cm
5. 0.5 cm

10.

An unstable atom with an excess of protons in the nucleus could be expected to decay by which of the following?

1. Alpha
2. Electron capture
3. Beta plus
4. Electron capture or beta plus
5. Electron capture, beta plus, or alpha

11.

Which of the following conditions must be met in order for positron emission to occur?

1. The nucleus must have an excess of neutrons.
2. The nucleus must have an excess of protons.
3. The nucleus must have an excess of neutrons and have an excess mass equal to two electrons.
4. The nucleus must have an excess of protons and have a -1 quantum state.
5. The nucleus must have an excess of protons and have an excess mass equal to two electrons.

12.

The exact mode of radioactive decay depends on all of the following EXCEPT:

1. Neutron to proton ratio is too high
2. Neutron to proton ratio is too low
3. The amount of excess energy of the parent
4. The atomic mass of the parent
5. The electron quantum state

13.

Alpha decay occurs when:

1. the parent is above 200 amu in mass.
2. the parent nucleus has excess neutrons.
3. the parent nucleus has excess protons.
4. nuclear force exceeds the binding energy.
5. the excess energy of the nucleus exceeds 5 MeV.

14.

Which effect predominates in the 10 keV to 100 keV region for photons?

1. Photoelectric (PE)
2. Compton
3. Pair production (PP)
4. Elastic scattering
5. Inelastic scattering

15.

Which effect predominates in the 100 keV to 5 MeV region for photons?

1. Photoelectric effect (PE)
2. Compton effect
3. Pair production (PP)
4. Elastic scattering
5. Inelastic scattering

16.

What effect predominates in the greater than 5 MeV ionizing region?

1. Photoelectric Effect (PE)
2. Compton effect
3. Pair production (PP)
4. Elastic scattering
5. Inelastic scattering

17.

KERMA is a unit of radiation measurement expressed in units of:

1. Joule per kilogram.
2. Joule per coulomb.
3. Coulomb per kilogram.
4. Disintegrations per kilogram.
5. Disintegrations per liter.

18.

In the beta particle decay of strontium the following occurs:

$^{90}Sr_{Z} \rightarrow {}^{A}Y_{38} + \beta^{-}$

What is the "A" value for yttrium (Y)?

1. 91
2. 90
3. 88
4. 87
5. 86

19.

In the beta decay of Strontium the following occurs:

$^{90}Sr_{38} \rightarrow {}^{90}Y_{Z} + \beta^{-}$

What is the value "Z" for Yttrium (Y)?

1. 39
2. 38
3. 42
4. 37
5. 36

20.

Isotopes are defined as:

1. different radionuclides decaying by the same decay mode.
2. forms of the same element containing different numbers of neutrons.
3. forms of the same element containing different numbers of electrons.
4. topes that have a temperature of 32°F.
5. radioactive species containing the same number of neutrons PLUS electrons.

21.

If a radionuclide undergoes eight half-lives, what percentage of the original activity remains?

1. 0.15%
2. 0.4%
3. 0.78%
4. 1.5%
5. 12.5%

22.

What is the "threshold" of radiation dose effects?

1. The dose of radiation that will kill any organism
2. The dose of radiation that will kill 50% of those exposed
3. The dose of radiation below which there are no effects whatsoever on the body
4. The dose of radiation that will begin to alter cell structure
5. The dose of radiation at which the damage effect is just balanced out by recovery

23.

Emission of a negatively charged beta particle results from the transformation of:

1. a neutron into a proton.
2. a proton into a neutron.
3. a neutrino into a neutron.
4. a positron into a proton.
5. an electron into a proton.

24.

Which of the following organ systems is the MOST radiosensitive?

1. Central nervous system
2. Bone marrow
3. Gastrointestinal system
4. Muscle system
5. Skeletal system

25.

"Below a certain radiation dose, no effects whatsoever occur in the human body." This statement supports which one of the following dose/effect theories?

1. Threshold
2. Linear Non-Threshold
3. Non-Linear Non-Threshold
4. Linear Quadratic
5. The "Hormesis" Effect

26.

If the maximum beta particle energy for a radionuclide is 7.0 MeV, the average beta particle energy is approximately:

1. 2.33 MeV
2. 0.223 MeV
3. 1.43 MeV
4. 0.256 MeV
5. 700 keV

27.

The mass defect is equal to:

1. the difference in binding energies between parent and daughter nuclides.
2. the difference between the mass of the nucleus as whole and the sum of the component nucleus masses.
3. the rest mass of particulate radiations emitted by the daughter nuclide.
4. the sum of the binding energies of the parent and daughter nuclides.
5. the rest mass energy equivalence of the daughter nuclide.

28.

In the energy range from 0.1 to 5 MeV in either air or water, the most predominant photon interaction is:

1. pair production.
2. Compton scattering.
3. photoelectric effect.
4. Rayleigh scattering.
5. Thomson scattering.

29.

Charged particles interact with matter by which of the following mechanisms?

1. Photoelectric effect and Compton scattering
2. Inelastic collision
3. Elastic scattering
4. Excitation and ionization
5. Bremsstrahlung and pair production

30.

The range of an alpha particle in tissue is approximately 50 to 100:

1. centimeters.
2. millimeters.
3. microns.
4. angstroms.
5. barns.

31.

After four half-lives, the original activity of a radionuclide will be reduced by a factor of:

1. 4
2. 8
3. 16
4. 32
5. 64

32.

A nuclide that undergoes orbital electron capture:

1. emits an electron, neutrino, and the characteristic x-rays of daughters.
2. emits a neutrino and the characteristic x-rays of daughters.
3. also decays by positron emission.
4. also emits internal conversion electrons.
5. makes an isomeric transformation.

33.

In the interaction of thermal neutrons with tissue, the major dose is from:

1. elastic scattering by hydrogen atoms.
2. the N-14 (n,p) C-14 reaction.
3. inelastic scattering by hydrogen atoms.
4. the H-1 (n, gamma) H-2 reaction.
5. the H-2 (n, gamma) H-3 reaction.

34.

Which of the following is NOT true for non-stochastic effects?

1. The severity of effect varies with dose
2. The probability of an effect occurring is considered a function of dose, without threshold
3. Risk factors are expressed per unit of dose (gray)
4. They will be strictly somatic effects
5. Examples are lens cataracts and impaired fertility

35.

The roentgen is equal to:

1. 1.0 coulomb/kg
2. 1.00×10^{-3} coulomb/kg
3. 5.28×10^{-3} coulomb/kg
4. 2.58×10^{-4} coulomb/kg
5. 5.28×10^{-4} coulomb/kg

36.

The LET:

1. refers to the energy lost by the charged particles per unit path traveled by the particle.
2. refers to the energy deposited in the medium by collision per unit path traveled by the charged particle in the medium.
3. includes both collision and radiation energy losses by the charged particle.
4. has the same value for neutrons and gamma rays.
5. is not related to quality factor.

37.

One Roentgen of gamma radiation is equivalent to absorbed doses in air and tissue of about how many rads, respectively?

1. 0.80 and 0.98 rads
2. 0.98 and 0.80 rads
3. 0.87 and 0.98 rads
4. 0.98 and 0.87 rads
5. 0.78 and 0.98 rads

38.

Radioactive atoms that have a neutron to proton ratio that is too large tend to decay by:

1. beta plus decay.
2. beta minus decay.
3. electron capture decay.
4. alpha decay.
5. isomeric transition.

39.

The number of ion pairs produced by photon interactions in a given small volume of air under electronic equilibrium conditions is a measure of the:

1. stopping power.
2. linear energy transfer.
3. specific ionization.
4. exposure.
5. absorbed dose.

40.

In living tissue, fast neutrons lose from 80 to 95 percent of their energy by interacting with:

1. hydrogen.
2. carbon.
3. nitrogen.
4. oxygen.
5. sodium.

41.

If the mean life of a radionuclide is 10 days, what is the half life?

1. 8
2. 14
3. 10
4. 5
5. 7

42.

Alpha decay can be described as the emission of a(n) __________________ from the nucleus.

1. helium nucleus
2. negative electron
3. positron
4. gamma ray
5. proton

43.

The most common type(s) of interaction(s) for beta particles is (are):

1. ionization only.
2. ionization, excitation, and bremsstrahlung.
3. photoelectric effect only.
4. photoelectric effect, Compton effect, and pair production.
5. pair production and excitation.

44.

The type of neutron interaction that is most important in moderating neutrons to thermal energies is:

1. elastic scattering.
2. inelastic scattering.
3. radiative capture.
4. fission.
5. fusion.

45.

The LD $_{50/30}$ for man from an acute, whole-body absorbed dose of penetrating gamma radiation is believed to be about:

1. 4 rad.
2. 40 rad.
3. 400 rad.
4. 4,000 rad.
5. 40,000 rad.

46.

The most probable process for energy deposition by a 1 MeV photon in tissue is:

1. photoelectric absorption.
2. pair production.
3. Compton scattering.
4. photonuclear absorption.
5. bremsstrahlung.

47.

Which of the following types of decay "competes" with positron decay?

1. Auger electrons
2. Internal conversion electrons
3. Beta decay
4. Electron capture
5. X-ray emission

48.

How many centimeters of lead were required for a shield if seven half-value layers were used to shield a beam of gamma photons? (attenuation coefficient for lead = 0.4559 per centimeter)

A. 1.52 cm
B. 10.6 cm
C. 13.2 cm
D. 15.2 cm
E. 30.4 cm

49.

What immediate action should be taken if skin is accidentally broken while working with radioactive substances?

A. Immerse the wound in tepid water.
B. Wash the wound under running water.
C. Apply a liberal portion of titanium dioxide paste.
D. Scrub with a brush using heavy lather and tepid water.
E. Wash the wound vigorously with a damp cloth.

50.

The basic physical methods applied to protection against external radiation hazards are:

A. film badges and dosimeters.

B. protective clothing.

C. time, distance, and shielding.

D. whole body counting and bioassay.

E. G-M survey meters.

51.

Leak test of sealed radioactive sources should be sensitive enough to detect:

A. 0.005 µCi.

B. 0.1 µCi.

C. 0.5 µCi.

D. 100 dpm/100 cm^2.

E. 1000 dpm/100 cm^2.

52.

Which of the following is in the order of MOST effective to LEAST effective shielding materials for gamma radiation sources?

A. Water, lead, concrete, iron

B. Water, lead, iron, concrete

C. Lead, water, iron, concrete

D. Lead, iron, concrete, water

E. Lead, iron, water, concrete

53.

If a shielding material has a half-value layer of 0.5 inches and a buildup factor of 2, how much shielding will be required to reduce the exposure rate from 200 milliroentgen per hour to 50 milliroentgen per hour?

- A. 0.5 inches
- B. 1.0 inches
- C. 1.5 inches
- D. 2.0 inches
- E. 3.0 inches

54.

The units used to express μ/ρ (mu over rho) are:

- A. 1/cm
- B. g/cm^2
- C. g/cm^3
- D. cm^2/g
- E. cm^3/g

55.

A photon beam is reduced to 0.1 of its initial intensity by lead with a linear attenuation coefficient of 0.567/cm. What is the shielding thickness?

- A. 4.06 cm
- B. 2.30 cm
- C. 1.0 cm
- D. 0.567 cm
- E. 0.05 cm

56.

A 500 millicurie Phosphorous-32 solution has spilled in a hospital room. A quick method of shielding the spill and preventing the spread of contamination would be to:

- A. open the window to completely ventilate the room.
- B. increase the building ventilation flow to dilute the solution throughout the building.
- C. use a broom to consolidate the spill and cover with towels.
- D. cover the spill with a plastic bed sheet.
- E. hose down the room and direct the spill to the floor drain system.

57.

You enter a room and are directing cleanup of a liquid spill. What precautions do you prescribe to prevent the spread of contamination?

- A. Clean the edges of the spill first.
- B. Install shielding around the spill.
- C. Work rapidly to minimize the dose.
- D. Install ventilation to assist in the cleanup of the spill.
- E. Dilute with large volumes of water.

58.

ICRP Publication 26 states, "No practice shall be adopted unless its introduction produces a net benefit." This concept is called which of the following?

- A. Optimization
- B. Maximization
- C. Minimization
- D. Justification
- E. Realization

59.

There has been a severe radiation accident at a Co-60 irradiation facility. The primary duty of the first health physicist to arrive on the scene is to:

1. arrange for analysis of personnel monitoring equipment.
2. arrange for hospitalization of the victims.
3. secure the facility in such a manner that the accident cannot immediately reoccur.
4. report the occurrence of the accident at the facility to the NRC by telephone.
5. reassure the victims to prevent psychogenic vomiting.

60.

All of the following apply to the term "transport index" EXCEPT:

1. Numerically is the highest radiation dose rate, in millirems per hour, at one meter from any accessible external surface of the package
2. Numerically is for Fissile Class II packages only, the number obtained by dividing the number "50" by the number of similar packages that may be transported together
3. Numerically is limited to a maximum of 50 per vehicle (other than a sole use vehicle)
4. Has the same limits for all types of carriers
5. Numerically is limited to a maximum of 10 per package (when shipped non-exclusive use)

61.

ICRP Publication 26 states, "No practice shall be adopted unless its introduction produces a net benefit." This concept is called:

1. optimization.
2. maximization.
3. minimization.
4. justification.
5. realization.

62.

Leak tests of sealed radioactive sources should be sensitive enough to detect which of the following?

1. 1000 dpm/100 cm^2
2. 100 dpm/100 cm^2
3. 0.1 μCi
4. 0.005 μCi
5. 0.5 μCi/cm^2

63.

ICRP 26 recommends the use of a weighting factor, w(T) to express the proportion of stochastic risk from an irradiated tissue, T, to the total risk when a body is uniformly irradiated. For the lung, this factor has a value of:

1. 0.03
2. 0.12
3. 0.25
4. 0.30
5. 0.50

64.

The safe transportation of radioactive materials is enhanced most by:

1. labeling requirements.
2. the use of special vehicles.
3. the actions of persons with special training to handle and transport the materials.
4. using labeling, packaging, and transportation methods commensurate with the quantity, type, and hazard of radioactive material.
5. forbidding the transportation of highly hazardous radioactive materials over bridges, through tunnels, and through densely populated cities.

65.

The maximum allowable radiation level at one meter from the surface of a package shipped in other than an exclusive-use vehicle is:

1. 1 mrem/hr
2. 25 mrem/hr
3. 10 mrem/hr
4. 100 mrem/hr
5. 200 mrem/hr

66.

Leak testing of sealed radioactive sources used for radiography should be performed:

1. prior to use of the device.
2. prior to and following transportation of the device.
3. prior to and following transportation of the device and prior to the use of the device.
4. every six months.
5. annually and prior to use of the device.

67.

At what level does loose alpha contamination become unacceptable on a radioactive materials package to be transported in a non-exclusive use load?

1. 10^{-5} μCi/cm^2
2. 220 dpm/cm^2
3. 2200 dpm/cm^2
4. 10^{-5} μCi
5. 10^{-4} μCi/cm^2

68.

The maximum allowable radiation level at contact of a package shipped in other than an exclusive use vehicle is:

1. 1 mR/hr
2. 2.5 mR/hr
3. 200 mR/hr
4. 500 mR/hr
5. 1000 mR/hr

69.

The use of a Radioactive Yellow III label on a package containing radioactive material, shipped non-exclusive use, implies:

1. that the radiation exposure rate from any point on the surface of the package does not exceed 0.5 mR/hr.
2. that the package contains Fissile Class II radioactive material and the transport index does not exce 0.5.
3. that the radiation exposure rate from any point on the surface of the package does not exceed 200 mR/hr and the transport index does not exceed 10.
4. that the radiation exposure rate from any point on the surface of the package does not exceed 10 mR/hr and that the transport index does not exceed 0.5.
5. that the package contains Fissile Class II radioactive material for which the transport index exceeds 10.

70.

According to ICRP Publication 60, the "equivalent dose" to a tissue is the product of the absorbed dose averaged over the tissue and:

1. an appropriate quality factor.
2. an appropriate radiation weighting factor.
3. an appropriate tissue weighting factor.
4. an appropriate nonstochastic risk coefficient.
5. an appropriate stochastic risk coefficient.

71.

According to 10 CFR 20 radiological surveys should be performed to evaluate which of the following?

1. The extent of the radiation levels
2. The concentrations of radioactive materials
3. The quantities of radioactive materials
4. The potential radiological hazards that could be present
5. All of the above

72.

A technician surveys a shielded pure beta-emitting source and obtains a measurement of 100 mR/hr. If there is adequate shielding present to shield all beta particles from the source, the reading is caused by:

1. Compton scattering in the shield material.
2. pair production in the shield material.
3. photoelectric effect in the shield material.
4. bremsstrahlung.
5. Cerenkov photons.

73.

Which of the following are considered as risks of chronic low-level exposure?

I. Sterility
II. Leukemia
III. Cataracts
IV. Skin erythema

1. I
2. II
3. III
4. I and III
5. II, III, and IV

74.

If a large group of people receive an acute deep dose of 450 rads, what fraction of the people would you expect to be alive in 2 months if they receive no medical treatment?

1. < 25%
2. 25%
3. 50%
4. 75%
5. 100%

75.

The primary mechanisms whereby charged particles lose energy in traversing materials are:

1. nuclear interactions involving ejection of multiple particles from the "target" nucleus.
2. bremsstrahlung effects arising in the electronic structures of the material.
3. ionization and excitation affecting the electronic structures of the material atoms.
4. gravitation interactions.
5. pair production interactions.

76.

You enter a room and are directing cleanup of a liquid spill. What precautions do you prescribe to prevent the spread of contamination?

1. Clean the edges of the spill first.
2. Install shielding around the spill.
3. Work rapidly to minimize the dose.
4. Install ventilation to assist the cleanup of the spill.
5. Dilute with large volumes of water.

77.

A Pu-Be source should be stored in a:

1. secured area and surrounded with sufficient lead shielding to reduce the exposure rate to permissible levels.
2. properly posted and secured area and surrounded with sufficient hydrogenous shielding to reduce the dose equivalent rate to permissible levels.
3. safe.
4. locked room posted with "Caution: High Radiation Area" signs.
5. locked room posted with "Caution: Neutron Radiation Hazard".

78.

In the shielding equation $I = I_0 e^{-\mu x}$, if μ is larger it means that the shielding material for the photon energy of interest is which of the following?

1. A more effective shield
2. A less effective shield
3. A material with undetermined shielding properties
4. No more effective than if μ was smaller
5. None of the above

79.

Which of the following is the MOST desirable method to provide radiological protection during maintenance on a highly contaminated component?

1. Ensure workers wear protective clothing and respirators.
2. Always use auxiliary ventilation.
3. Attempt to remove contamination prior to maintenance.
4. Ensure workers are properly trained in maintenance.
5. Cover any exposed skin surfaces on workers.

80.

The buildup factor for a gamma source and shield geometry at a point outside the shield is 20. The fraction of the dose contributed by unscattered photons is:

1. 0.05
2. 0.1
3. 0.5
4. 0.8
5. 0.9

81.

How are tools or equipment transferred from a contaminated area?

1. Sealed in clean plastic bags
2. Decontaminated thoroughly before being transferred to another contaminated area
3. Painted
4. Sealed in aluminum boxes
5. Immersed in heavy water

82.

The basic physical methods applied to protection against external radiation hazards are:

1. film badges and dosimeters.
2. protective clothing.
3. time, distance, and shielding.
4. whole body counting and bioassay.
5. G-M survey meters.

83.

Which of the following are distinguishing colors and symbol for radiation warning signs?

1. Yellow and magenta with a trifoil
2. A yellow and magenta atom
3. A black and yellow atom
4. A solid yellow trifoil on black background
5. A red and white molecule

84.

In planning shielding for walls of an x-ray room that are not subjected to the direct beam, consideration is given to provide shielding for:

1. scattered radiation from all sources and leakage from the x-ray tube.
2. scattered radiation from natural background.
3. scattered radiation from the patient only.
4. leakage radiation from the x-ray tube.
5. secondary scattered radiation from the walls.

85.

The linear attenuation coefficient of a shielding material for gamma photons varies depending on the density of the shielding material and the:

1. mass of the shielding material.
2. energy of the photon.
3. photon yield.
4. half-life of the isotope.
5. mean-life of the isotope.

86.

For removing radioactive contamination from the skin, your best choice would be:

1. a mild soap.
2. a dilute $KMnO_4$ solution.
3. a dilute H_2SO_4 solution.
4. an acetone-alcohol solution.
5. a strong caustic solution.

87.

An ionization chamber has voltage great enough to cause:

1. ions before recombination; no secondary electrons.
2. secondary electrons with a short range avalanche.
3. secondary electrons with avalanche to whole anode.
4. continuous discharge.
5. recombination before current is produced.

88.

A proportional counter has enough voltage to cause:

1. ionization before recombination; no secondary electrons.
2. secondary electrons with short range avalanche.
3. secondary electrons with long range avalanche.
4. continuous discharge.
5. recombination prior to pulse.

89.

A Geiger counter has enough voltage to cause:

1. electrons before recombination; no secondary electrons.
2. secondary electrons with a short range avalanche.
3. secondary electrons with a long range avalanche.
4. continuous discharge.
5. recombination prior to pulse.

90.

Ionization chambers have gas amplification factors:

1. equal to 1.
2. greater than 1.
3. less than 1.
4. dependent on the voltage.
5. dependent on the volume.

91.

Proportional counters have gas amplification factors that are:

1. equal to 1.
2. greater than 1.
3. less than 1.
4. independent of the voltage.
5. dependent on the volume.

92.

Geiger counters have gas amplification factors:

1. equal to 1.
2. greater than 1.
3. less than 1.
4. dependent on the voltage.
5. dependent on the volume.

93.

The average energy required to produce an ion pair in air by x-ray or gamma radiation is:

1. 16 eV
2. 32 eV
3. 32.5 eV
4. 33.7 eV
5. 35 eV

94.

A classic type of interaction occurs when a positron and an electron interact to produce 2 photons each with an energy of 511 KeV. The reaction is called:

A. Compton effect.

B. Annihilation.

C. Fusion.

D. Electron fission.

E. Pair production.

95.

One of the processes that increases the neutron/proton ratio in a nucleus involves the conversion of a proton to a neutron. This process is called:

A. Beta minus decay

B. Electron capture

C. Fission

D Fusion

E Isomeric transition

96.

Radioactive decay is said to be first order decay. It is also ------------ temperature.

A. dependent on

B. independent on

B. dependent on density

C. dependent on entropy

D. dependent on pressure

97.

If I-131 has a biological half-life of 138 days, what Is the effective half-life of l-131?

A. 7.6 d

B. 8.05 d

C. 59.4 d

D. 41.25 d

98.

Gamma radiation produces ionization by which of the following?

A. 'Photoelectric effect . Compton effect , pair production

B. Photoelectric effect, Compton effect. bremsstrahlung

C. Bremsstrahlung, photoelectric effect

D. Excitation, photoelectric effect, pair production

E. Excitation and bremsstrahlung,